

Project Idea:

MediScript AI

Intelligent Medical Prescription Interpretation, Translation and Patient Assistance System

1. Project Overview

Project Name: MediScript AI

Positioning Statement: An AI-powered patient assistance platform that transforms difficult-to-read medical prescriptions into understandable, personalized, and interactive guidance while keeping the prescribing doctor's instructions as the source of truth.

Core Philosophy: MediScript AI bridges the gap between complex medical documents and patient comprehension. It strictly avoids replacing medical professionals; instead, it acts as an intelligent translator and organizer, using Retrieval-Augmented Generation (RAG) to ground all information in trusted medical references.

2. Problem Statement

Medical prescriptions are notoriously difficult for patients to understand due to poor handwriting, Latin abbreviations (e.g., SOS, OD, BD), complex dosage timing, and language barriers. This lack of comprehension leads to medication non-adherence, accidental overdoses, and dangerous drug interactions. Furthermore, when patients seek explanations online, they risk encountering hallucinations from unverified AI tools or generic search results that do not account for their specific medical profiles.

3. Objectives

- **Accurate Extraction:** Accurately digitize handwritten and printed prescriptions using Vision-Language Models (VLMs) and Optical Character Recognition (OCR).
- **Semantic Understanding:** Convert medical shorthand into plain, accessible language.
- **Grounded Explanations:** Provide evidence-based medication information using RAG, preventing AI hallucinations.
- **Personalization:** Tailor guidance based on a user's specific health profile (allergies, chronic conditions).
- **Safety First:** Identify drug interactions and strictly route diagnostic or medical-decision queries back to healthcare professionals.

4. Target Users

- **Patients:** Seeking clear, multilingual, and personalized understanding of their treatments.
- **Caregivers:** Family members managing medications for elderly or dependent patients.
- **Pharmacists:** Requiring quick interpretation of illegible handwriting.
- **Hospitals/Clinics:** Seeking to improve patient engagement and post-visit medication adherence.

5. Scope & Safety Constraints

In Scope: Prescription digitization, dosage simplification, interaction checking, personalized reminders, multilingual translation, and RAG-based medical information retrieval.

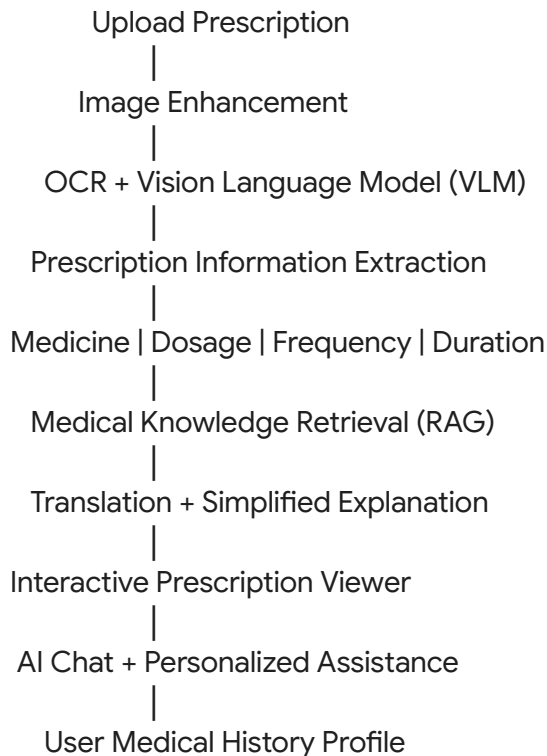
Out of Scope (Hard Boundaries): Medical diagnosis, disease prediction, altering prescribed dosages, and providing emergency medical advice.

5.1 Three-Tier Response Mode (Safety Guardrails)

To maintain clinical safety, the AI operates in three strict modes:

1. **Prescription Explanation:** Explains exactly what is written and what the dosage shorthand means (e.g., "1-0-1 means morning and night").
2. **Medication Information:** Provides evidence-based information retrieved *only* from trusted sources via RAG (e.g., "This is commonly used for allergies").
3. **Safety Guidance:** Identifies potential concerns and advises consulting a doctor. (e.g., ⚠️ *"Should I stop taking this antibiotic because I feel better?" → Explains general guidance to finish the course and advises contacting the prescribing doctor rather than deciding for the patient.*)

6. System Architecture



7. Core Modules

Module 1 — Prescription Upload & Pre-processing

Patients can upload via Image, PDF, or Camera photo. The system automatically pre-processes the image using OpenCV/Pillow (cropping, denoising, sharpening, contrast improvement) to prepare it for optimal OCR extraction.

Module 2 — Prescription Reading (The Engine)

Moves beyond plain OCR by combining traditional OCR (Tesseract/PaddleOCR), Vision Language Models (Florence-2/Qwen2.5-VL), and NLP to extract structured JSON data.

Example extraction:

```
{
  "medicine": "Amoxicillin",
  "strength": "500mg",
  "dosage": "Morning and Night",
  "duration": "5 days",
  "instruction": "After Food"
}
```

Module 3 — Medical Knowledge Retrieval (RAG)

Instead of relying on LLM internal memory, the system queries a verified vector database (FAISS/Qdrant) for drugs (e.g., "Levocetirizine"). It returns grounded facts: *Used for allergies, may cause drowsiness, avoid alcohol, take at night.*

Module 4 — Patient-Friendly Translation

Translates medical shorthand into intuitive UI elements:

- 1-0-1 → Morning  | Afternoon  | Night 
- SOS → *Take only when symptoms become severe.*
- OD → *Take once every day.*

Module 5 — Interactive Prescription

A clickable digital twin of the prescription. Clicking on "Paracetamol 650mg" opens an contextual chat where the AI answers questions (e.g., "Can I take this with milk?") using the uploaded prescription, RAG knowledge, and conversation history.

Module 6 — Personalized User Profile

Maintains user data (Age, Weight, Allergies, Chronic diseases, Current medicines).

Value Add: If a prescription contains Amoxicillin, but the profile lists a Penicillin allergy, the system flags: *"Your profile indicates a penicillin allergy. Contact the prescribing doctor before taking this."* (Verification, not diagnosis).

Module 7 — Medicine Timeline

A "Google Calendar for Medicines" visualizing daily intake schedules (e.g., 8 AM: ✓ Medicine A, 2 PM: ✓ Medicine B).

Module 8 — Medication Reminders

Push notifications or SMS alerts configured to the extracted schedule: *"Time to take Paracetamol 650mg. Take after food."*

Module 9 — Drug Interaction Detection

Cross-references newly prescribed medicines against the user's active profile and highlights known interactions, duplicate medicines, and warnings with cited references.

Module 10 — Explain Everything

An AI explainer that breaks down the *why* behind doctor instructions in simple terms (e.g., Why after food? Why a syrup instead of a tablet?).

Module 11 — Multi-language Support

Full translation of medicine names, schedules, and AI explanations into regional languages (Marathi, Hindi, Gujarati, Tamil, etc.).

Module 12 — Doctor Notes Simplifier

Translates clinical notes (e.g., "Review PRN or 7 days") to patient-friendly instructions ("Visit your doctor again after seven days, or sooner if needed").

Module 13 — Smart Search

Semantic search across current and past prescriptions. Typing "painkiller" highlights Paracetamol; typing "blood pressure" highlights Amlodipine.

Module 14 — Health Dashboard

A centralized hub tracking Current Medicines, Past Medicines, Completed Courses, Missed Medicines, Doctor Visits, and Downloaded Prescriptions.

8. Standout Differentiators (Advanced Features)

- **AI Confidence Scoring:** Displays system confidence metrics (e.g., Medicine Recognition: 98%, Dosage: 76%). If confidence drops below a threshold, the UI forces the user to manually verify the text.
- **Prescription Comparison:** Automatically diffs prescriptions from different visits, highlighting what was Removed, Added, or if the Dose/Duration changed.
- **Caregiver Mode:** Role-based access allowing family members to manage and monitor medication timelines for elderly patients while the patient retains ownership of the data.

9. Technology Stack

Component	Technology
Frontend	React / Next.js, TypeScript, Tailwind CSS, Framer Motion
Backend	FastAPI (Python) or Node.js
Database	PostgreSQL (Relational), Redis (Caching)
Vector DB	FAISS, Qdrant, or ChromaDB
Vision / OCR	Tesseract, PaddleOCR, TrOCR, Florence-2, Qwen2.5-VL
NLP Pipeline	spaCy, scispaCy, HuggingFace Transformers
RAG Framework	LangChain or LlamaIndex
LLM	GPT-4o, Gemini 1.5, or Llama 3 (Configurable)
Auth & Storage	JWT, OAuth, AWS S3 / Cloudinary
Deployment	Docker, NGINX, GitHub Actions, AWS / Vercel / Render

10. Non-Functional Requirements (NFRs)

- **Performance:** End-to-end OCR and extraction should process in under 10 seconds.
- **Reliability:** 100% reliance on RAG for medical facts (Zero-Hallucination tolerance for drug specs).
- **Security:** JWT authentication, Encrypted Storage (AES-256 for medical data at rest), HIPAA-compliant design considerations.
- **Accessibility:** WCAG 2.1 AA compliant frontend for elderly and visually impaired users.

11. Evaluation Metrics for Project Defense

To prove the system's engineering validity, the following metrics will be tracked:

- **OCR/Vision Accuracy:** Character Error Rate (CER) and Word Error Rate (WER) across diverse handwriting samples.
- **Extraction Accuracy:** Precision, Recall, and F1 Score for Medical Entity Recognition (Medicines, Dosages, Timings).

- **RAG Effectiveness:** Retriever Recall@K and Mean Reciprocal Rank (MRR).
- **LLM Safety:** Hallucination rate (must be near 0%) and Groundedness score (verifying output matches retrieved context).

12. Resume Highlights (For Post-Graduation)

- *Built an end-to-end AI healthcare platform integrating Vision-Language Models (VLMs), OCR, and Retrieval-Augmented Generation (RAG) to digitize and interpret complex medical prescriptions.*
- *Developed a robust extraction pipeline for medical entities (dosage, frequency, duration) and implemented a 3-tier semantic safety framework preventing AI hallucinations.*
- *Designed a scalable microservices architecture using FastAPI, Next.js, and PostgreSQL, incorporating vector embeddings (FAISS/Qdrant) for real-time drug interaction and contraindication checking based on personalized user health profiles.*